

# Evidencia — Reconocimiento de la red OT (nmap)

**Fase:** 3 — Demostración de ataques **Técnica MITRE ATT&CK for ICS:** T0846 — Remote System Discovery **Origen del escaneo:** contenedor kali (red red\_ot, 172.21.0.50) **Objetivo:** 172.21.0.0/24 **Comando:**

```
docker exec -it kali nmap -sV 172.21.0.0/24
```

**Fecha:** 21 de julio de 2026

## Resumen de hosts descubiertos

| IP          | Host              | Puerto             | Servicio detectado                                     | Observacion                                                                                                                       |
|-------------|-------------------|--------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 172.21.0.1  | Gateway de la red | —                  | —                                                      | Todos los puertos filtrados (sin respuesta)                                                                                       |
| 172.21.0.10 | openplc           | 8080/tcp           | Werkzeug httpd 0.16.0 (Python 2.7.17)                  | Interfaz web de gestión de OpenPLC. Puerto 502 (Modbus) aún visible — pendiente de activar el servidor Modbus en la configuración |
| 172.21.0.11 | conpot            | 2121/tcp, 8800/tcp | Sin fingerprint oficial (nmap no reconoce el servicio) | El honeypot responde imitando un <b>Siemens SIMATIC S7-200 (CP 443-1 EX40)</b> — ver nota abajo                                   |
| 172.21.0.20 | wazuh.manager     | —                  | —                                                      | Todos los puertos cerrados desde esta red (comportamiento esperado, Wazuh no expone servicios hacia OT)                           |

## Nota de interpretación: comportamiento del honeypot

Al recibir una petición HTTP no reconocida, Conpot devuelve una página de error 404 que se identifica explícitamente como un dispositivo industrial real:

```
<TITLE>Not found - Siemens, SIMATIC, S7-200</TITLE>
<h2>CP 443-1 EX40</h2>
```

```
Nmap scan report for openplc.tfmaquacontrol_red_ot (172.21.0.10)
Host is up (0.0000050s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
8080/tcp  open  http    Werkzeug httpd 0.16.0 (Python 2.7.17)
MAC Address: EA:C5:C3:75:78:01 (Unknown)
```

```
Nmap scan report for conpot.tfmaquacontrol_red_ot (172.21.0.11)
Host is up (0.0000050s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
2121/tcp  open  ccproxy-ftp?
8800/tcp  open  sunwebadmin?
2 services unrecognized despite returning data. If you know the
service/version, please submit the following fingerprints at
https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT
INDIVIDUALLY)=====
SF-Port2121-TCP:V=7.99%I=7%D=7/21%Time=6A5F967A%P=x86_64-pc-linux-
gnu%r(NU
SF:LL,17,"200\x20FTP\x20server\x20ready\.\r\n")%r(GenericLines,3A,"20
SF:FTP\x20server\x20ready\.\r\n500\x20Command\x20'\r\n'\x20not\x20
SF:tood\r\n")%r(GetRequest,39,"200\x20FTP\x20server\x20ready\.\r\n500
SF:ommand\x20'GET'\x20not\x20understood\r\n")%r(HTTPOptions,3D,"200\
SF:\x20server\x20ready\.\r\n500\x20Command\x20'OPTIONS'\x20not\x20unc
SF:od\r\n")%r(RTSPRequest,3D,"200\x20FTP\x20server\x20ready\.\r\n500\
SF:mmmand\x20'OPTIONS'\x20not\x20understood\r\n")%r(RPCCheck,32,"200\
SF:\x20server\x20ready\.\r\n501\x20can't\x20decode\x20command\.\r\n")
SF:SVersionBindReqTCP,95,"200\x20FTP\x20server\x20ready\.\r\n500\x20
SF:d\x20'\x00\x1e\x00\x06\x01\x00\x00\x01\x00\x00\x00\x00\x00
SF:\x00\x07VERSION\x04BIND\x00\x00\x10\x00\x03'\x20not\x20unc
SF:od\r\n")%r(DNSStatusRequestTCP,6E,"200\x20FTP\x20server\x20ready\
SF:00\x20Command\x20'\x00\x0c\x00\x00\x10\x00\x00\x00\x00\x00\
SF:00\x00\x00\x00'\x20not\x20understood\r\n")%r(Help,152,"200\x20F
SF:0server\x20ready\.\r\n214-
The\x20following\x20commands\x20are\x20recogn
```

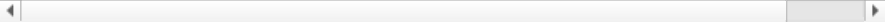
```

SF:ized:\r\n\x20'ABOR'\x20'ALLO'\x20'APPE'\x20'CDUP'\x20'CWD'\x20\x2f
SF:'\x20'HELP'\x20'LIST'\r\n\x20'MDTM'\x20'MKD'\x20\x20'MODE'\x20'NL$
SF:0'NOOP'\x20'PASS'\x20'PASV'\x20'PORT'\r\n\x20'PWD'\x20\x20'QUIT'\x
SF:IN'\x20'REST'\x20'RETR'\x20'RMD'\x20\x20'RNFR'\x20'RNT0'\r\n\x20'$
SF:x20'SIZE'\x20'STAT'\x20'STOR'\x20'STOU'\x20'STRU'\x20'SYST'\x20'T\
SF:\n\x20'USER'\r\n214\x20Help\x20command\x20successful\.\r\n")%r(SSL
SF:onReq,32,"200\x20FTP\x20server\x20ready\.\r\n501\x20can't\x20decod
SF:command\.\r\n")%r(TerminalServerCookie,32,"200\x20FTP\x20server\x2
SF:y\.\r\n501\x20can't\x20decode\x20command\.\r\n")%r(TLSSessionReq,3
SF:0\x20FTP\x20server\x20ready\.\r\n501\x20can't\x20decode\x20commanc
SF:n")%r(Kerberos,32,"200\x20FTP\x20server\x20ready\.\r\n501\x20can't
SF:ecode\x20command\.\r\n")%r(SMBProgNeg,32,"200\x20FTP\x20server\x2f
SF:\.\r\n501\x20can't\x20decode\x20command\.\r\n")%r(X11Probe,63,"20f
SF:TP\x20server\x20ready\.\r\n500\x20Command\x20'L\x00\x0b\x00\x0f
SF:0\x00\x00\x00\x00\x00\x00'\x20not\x20understood\r\n");
=====NEXT SERVICE FINGERPRINT (SUBMIT
INDIVIDUALLY)=====
SF:Port8800-TCP:V=7.99%I=7%D=7/21%Time=6A5F967A%P=x86_64-pc-linux-
gnu%r(Ge
SF:tRequest,7E,"HTTP/1.1\x20302\x20Found\r\nDate:\x20Tue,\x2021\x20:
SF:02026\x2015:49:38\x20GMT\r\nContent-
Type:\x20text/html\r\nLocation:\x20
SF:/index\.\html\r\nContent-
Length:\x200\r\n\r\n")%r(HTTPOptions,9B,"HTTP/1
SF:.1\x20200\x20OK\r\nDate:\x20Tue,\x2021\x20Jul\x202026\x2015:49:38
SF:MT\r\nAllow:\x20GET,HEAD,POST,OPTIONS,TRACE\r\nContent-
Length:\x200\r\n
SF:Connection:\x20close\r\nContent-
Type:\x20text/html\r\n\r\n")%r(FourOhFo
SF:urRequest,13C,"HTTP/1.1\x20404\x20Not\x20Found\r\nDate:\x20Tue,\x
SF:x20Jul\x202026\x2015:49:38\x20GMT\r\nContent-
Length:\x20232\r\n\r\n<HTM
SF:L>\n\n\x20\x20\x20\x20<HEAD>\n\x20\x20\x20\x20\x20\x20\x20\x20<TI
SF:t\x20found\x20-\x20Siemens,\x20SIMATIC,\x20S7-
200</TITLE>\n\x20\x20\x20
SF:\x20</HEAD>\n\n\x20\x20\x20\x20<BODY>\n\x20\x20\x20\x20\x20\x20\x2
SF:<h2>CP\x20443-
1\x20EX40</h2>\n\x20\x20\x20\x20\x20\x20\x20\x20<hr>\n\x2
SF:0\x20\x20\x20\x20\x20\x20\x20\x20&nbsp;
<br>\n\x20\x20\x20\x20\x20\x20\x20\x20\x20\x
SF:20This\x20resource\x20could\x20not\x20be\x20found\
<br>\n\x20\x20\x20\x20\x
SF:20</BODY>\n\n</HTML>");
MAC Address: BE:A9:FD:AE:D2:FE (Unknown)

Nmap scan report for wazuh.manager.tfmaquacontrol_red_ot
(172.21.0.20)
Host is up (0.000020s latency).
All 1000 scanned ports on wazuh.manager.tfmaquacontrol_red_ot
(172.21.0.20) are in ignored states.

```

Not shown: 1000 closed tcp ports (reset)  
MAC Address: AE:41:AF:A6:32:A9 (Unknown)



## Pendiente para el siguiente paso

El puerto 502 (Modbus TCP) de OpenPLC no aparece todavía porque el servidor Modbus no está activado ni hay un programa de control cargado. Antes de repetir este escaneo:

1. Activar “Enable Modbus Server” en Settings de OpenPLC (puerto 502).
2. Subir un programa de control (ej. válvula/bomba en Structured Text).
3. Repetir `nmap -sV 172.21.0.10` para confirmar el puerto 502 abierto — este será el vector de ataque de la siguiente fase (T0855 — Unauthorized Command Message).